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# The UV-divergence problem in statistical seismology: insights from an ETAS model with smoothed minimum triggering magnitude

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**The seismic fertility of small earthquakes to trigger earthquakes is still debated. If small earthquakes follow the same fertility scaling law as large earthquakes, their cumulative contribution leads to a divergence in the Epidemic-Type Aftershock Sequence (ETAS) model, which is traditionally addressed by introducing an ultraviolet (UV) cut-off—a minimum magnitude $m_0$ below which earthquakes are assumed to be infertile and do not trigger further events. We introduce a refined version of the ETAS model to rigorously constrain and estimate $m_0$, even below the completeness magnitude. By analyzing network-based and AI-enhanced earthquake catalogs and conducting simulations, we find that $m_0$ lies below the detection threshold of the best current catalogs.**

## INTRODUCTION

Earthquakes inherently trigger other earthquakes, a process central to the spatio-temporal organization of seismicity. When an earthquake occurs, stress redistributes in the brittle lithosphere, promoting aftershocks in areas of increased stress while inhibiting them in stress shadows. The Gutenberg-Richter law describes the distribution of earthquake magnitudes as an exponential function, resulting in cascades of nearly infinitely many small-energy events following any quake. This poses the ultraviolet (UV) divergence problem, where the triggering potential of these tiny events may not diminish sufficiently fast with decreasing magnitude, potentially escalating into large collective triggering intensities due to the smallest earthquakes. Additionally, undetected seismicity complicates the estimation of key physical metrics, such as spatio-temporal clustering and the branching ratio (Sornette and Werner, 2005). In the standard ETAS model of statistical seismicity, the ultraviolet (UV) problem is addressed by assuming the existence of a minimum triggering magnitude, $m_0$, below which earthquakes are considered too small to trigger others. Despite its critical physical implications, no research has been able to test for the existence and measure $m_0$ reliably (see however Li et al., 2024 for a new technique in this direction).

## OPEN QUESTIONS & POSSIBLE ANSWERS

- How can we reliably estimate the minimum triggering magnitude?
- How small is $m_0$?

✓ We can estimate $m_0$ by modifying the ETAS model to include the effect of the expected disappearance of the triggering power below a threshold magnitude.
✓ $m_0$ is found smaller than the completeness magnitude of both network- and AI-based catalogs.

## SIGNIFICANCE

If $m_0$ does not exist, forecasting earthquake occurrence from past seismic data becomes fundamentally unreliable, as every stress perturbation could potentially cascade into an avalanche leading to a mainshock. Moreover, the existence and value of $m_0$ should provide better understanding in the physics of earthquake nucleation and triggering as well as insights in the possible existence of characteristic scales.

## METHODS

We implement an ETAS model incorporating magnitude-dependent triggering power to test for the presence of cut-offs and to estimate them. We propose a new variant of the ETAS model with a S-shaped threshold $m_0$ given by

$$f_{m_0}(m-m_0) = \frac{e^{\delta(m-m_0)}}{1+e^{\delta(m-m_0)}}, \qquad (1)$$

where $\delta$ is inversely proportional to the width of the transition region between the two different regimes; the inflection point is $m_0$.

Our choice reflects the realistic assumption that the ability to trigger earthquakes decreases gradually as the size of the perturbing event becomes smaller. Figure 1 on the right represents the sigmoid function $f_{m_0}(m-m_0)$. We profile $\delta$ as a function of different possible values of $m_0$ in a wide range of magnitudes looking for its maximum; we compare our results with simulations in order to identify possible sources of bias and spurious results as well as to find clear markers to highlight $m_0$ in seismic catalogs. We also investigate the relationship of $\delta$ with other quantities of interest such as the branching ratio (Figure 2H).

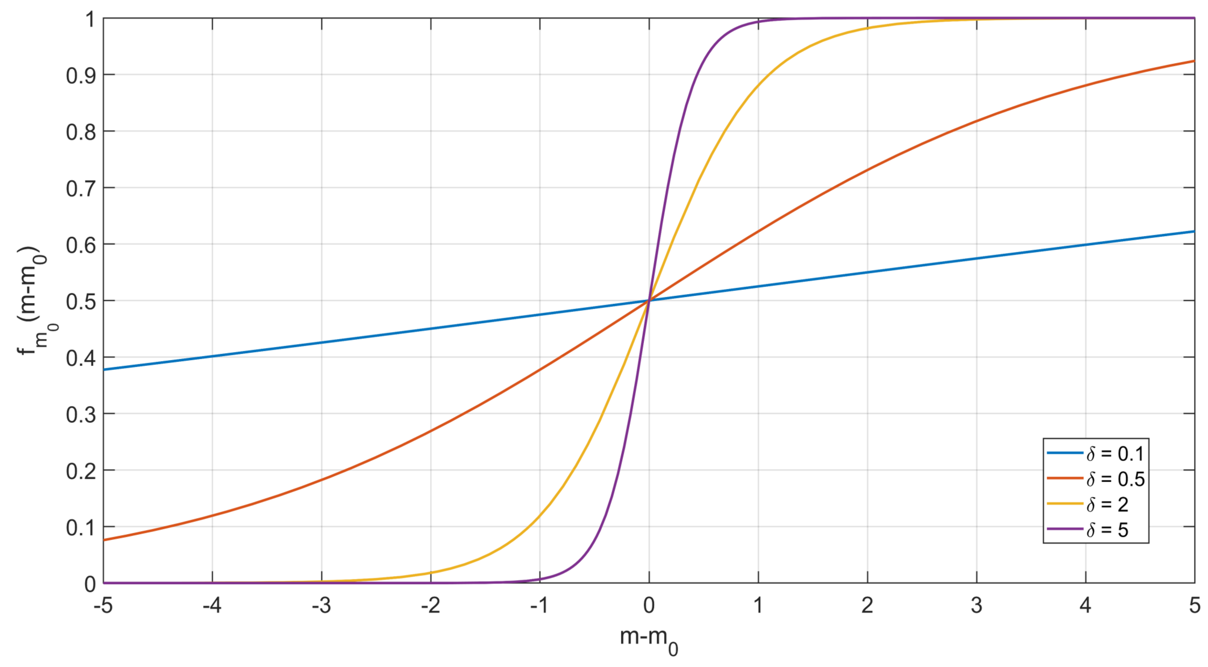



**Figure 1: The kernel of the ETAS model $g(x,y,m,t)$ is modified according to the transformation $g(x,y,m,t) = f_{m_0}(m-m_0)\, g(x,y,m,t)$, where $f_{m_0}(m-m_0)$ is the sigmoid function (Eq. 1) above.**

## RESULTS

Our statistical calibration of various datasets (here, as an example, we report results for the China Seismic Experimental Site CSES Catalog, 2000-2024 - Figure 2A, B, E - and the Central Italy AI-enhanced 2016-2017 catalog by Tan et al., 2021) generally finds very small values of $\delta$ indicating that $m_0$ lies below the typical magnitude of completeness of both ML-augmented and instrumental in seismic catalogs. Simulations show that an increase of $\delta$ is expected in the range of magnitudes around the minimum triggering one with a narrower peak for higher delta values (Figure 2C). $m_0$ is expected to be detectable even though below $m_c$ if their difference is not large (e.g., Figure 2D). Conversely, no peak of $\delta$ can be detected if the minimum triggering magnitude is much smaller than the completeness magnitude (Figure 2G).

We identify three main sources of bias in the estimation of $\delta$ (in both simulated and real seismic catalogues):

**1) The duration of the catalog:** short catalogues lead to higher estimated $\delta$. See Figure 2E. A possible explanation is that short-term incompleteness may produce an upward bias in the estimation of $\delta$ if the catalog is only a few years long or shorter.

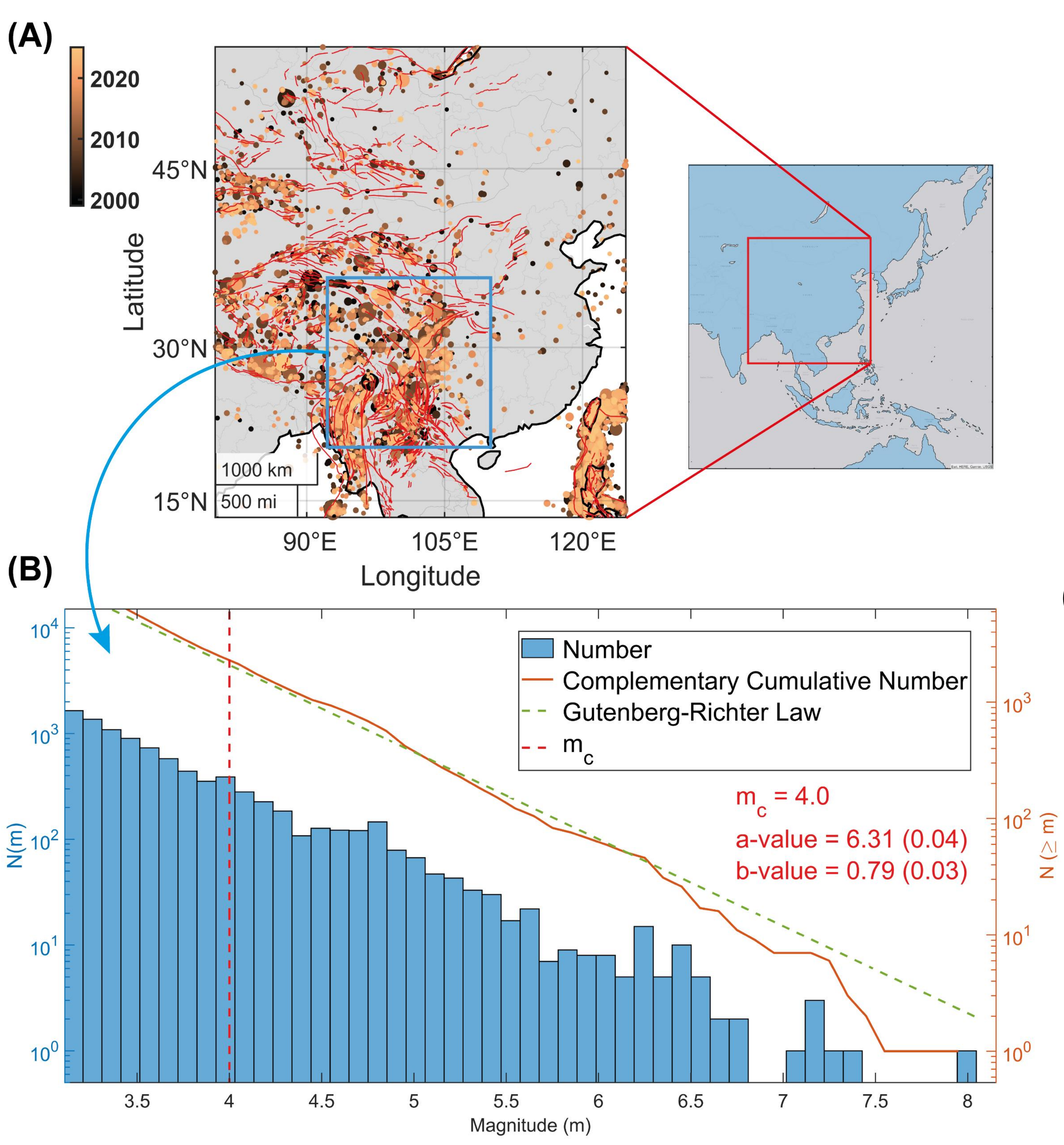
**2) Catalog incompleteness:** $\delta$ is higher close to the completeness magnitude likely due to an increasing percentage of missing events while approaching $m_c$. See Figure 2C, D, E, F.

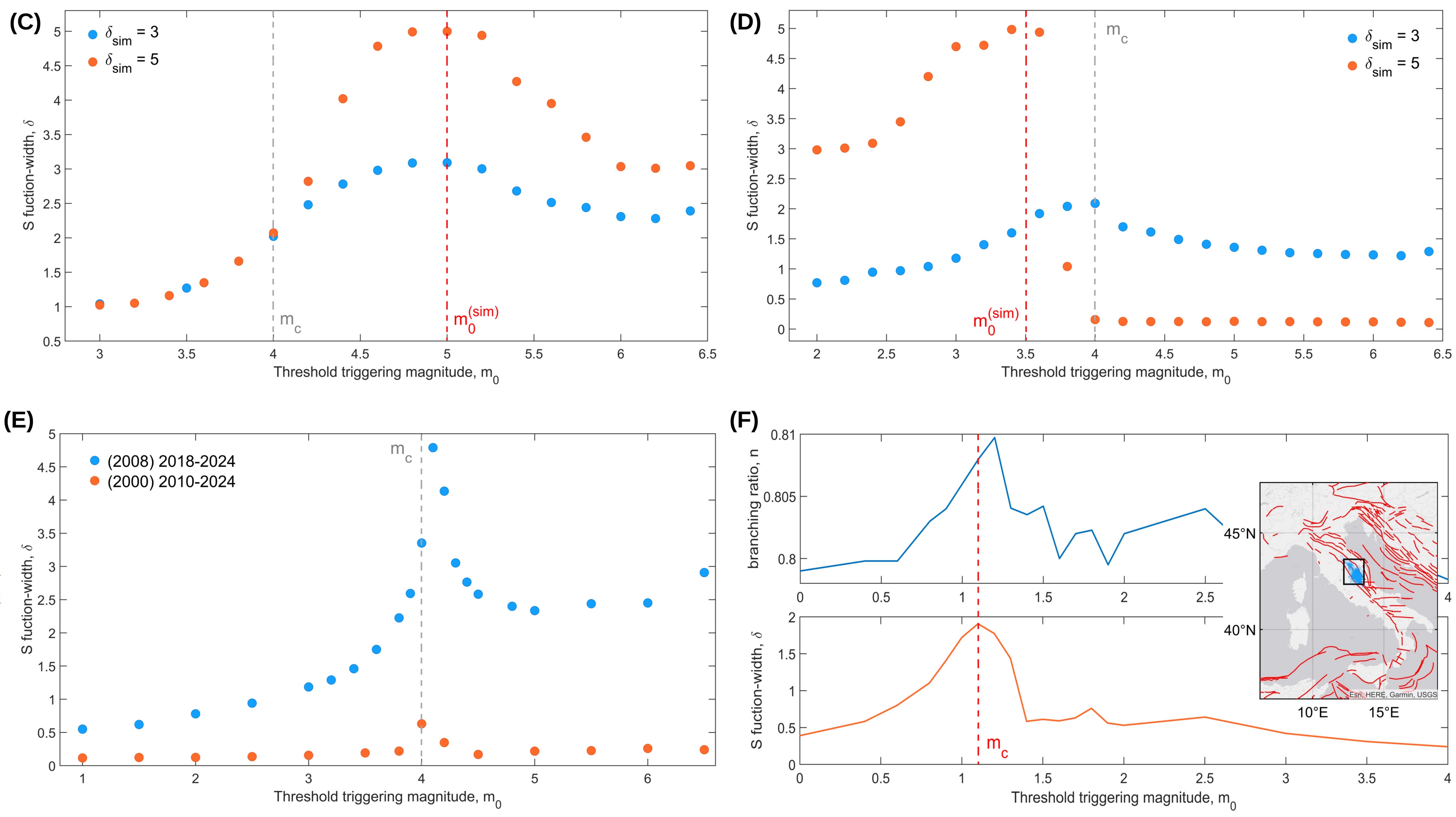
**3) Analysis in the large magnitudes range:** $\delta$ slightly increases profiling $m_0$ at large magnitudes ($m_0 > m_{max} - 2$, where $m_{max}$ represents the maximum magnitude in catalog). See the right side of Figure 2C, D, E.

## CONCLUSIONS

Our findings suggest that much of the underlying seismicity dynamics responsible for detected seismic events remains unobservable. Earthquakes that cannot be detected by existing networks may contribute to triggering future earthquakes of possibly large magnitudes. Significant values of $\delta$ ($\delta \geq 1$) are mostly associated with catalog incompleteness. Analogous results are also found when investigating instrumental seismic catalogs available in Southern California, Italy, Iceland, New Zealand and Türkiye.

**Figure 2: (A) Map of the CSES Catalog (m > 4, 2000-2024), (B) completeness and frequency magnitude distribution of the catalog. (C) and (D) represent the profile of $\delta$ simulated for the CSES Catalog with $m_0$ = 5.0 and $m_0$ = 3.5 respectively. The peak of $\delta$ is located around $m_0$ as expected even in the case $m_0 < m_c$ = 4; however, also the effect of catalog incompleteness is detected in the latter one. It is coherent with the profiling of $\delta$ shown in (E), where $\delta$ is larger for the shorter catalog. (F) represents the analysis of the branching ratio and $\delta$ as a funtiion of $m_0$ in the case of the high-quality ML-enhanced catalog in Central Italy (2016-2017). Even in this case, the minimum triggering magnitude is too small to be constrained; the peak of $\delta$ is due to catalog incompleteness. The branching ratio and $\delta$ are positively correlated (H). (G) is a simulation of the CSES catalog with a threshold triggering magnitude set at $m_0$ = 0 ($m_c - m_0 \gg 1$): the calibration procedure is not able to detect it.**

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